

Computer Engineer with proficient experience developing software solutions in both industry and academia. Proficient in front-end technologies (HTML, CSS, JavaScript) and back-end development (SQL, API, Java), with a strong background in Python and C++ gained through robotics research at the University of Cape Town. I am motivated to create scalable, intelligent software solutions that address real-world challenges and deliver meaningful social impact.

SKILLS

Programming Languages	Python, C++, Java, MATLAB
Front-End	HTML, CSS, JavaScript
Database	MySQL, PySpark, SparkSQL, APIs, Flask, Azure Databricks, Power BI, Tableau
Development Tools	Git, Linux, ROS2, Shell Scripting, Jupyter/Google Colab, AWS, IDEs
Software Concepts	Object-Oriented Programming (OOP), Multithreading, Unit Testing, Agile/Scrum

EDUCATION

Bachelor of Science in Electrical and Computer Engineering, *University of Cape Town* **2019 — 2022**

Awards: Dean's Merit List, UCT Plus Bronze Leadership Award, Engineering Mentorship Award

Key Courses: Introduction to Python and Java, C++ and Machine Learning, Data Structures and Databases, Concurrent Programming (Java), Digital Signal Processing, Embedded Systems, Intro to Medical Engineering, Law for Engineers, Managerial Finance, Social Impact of Engineering

Master of Science in Engineering (Mechatronics), *University of Cape Town* **2023 — 2025**

Research: Vision-Only Socially Compliant Navigation Robotics

Key Skills: Python, C++, Deep Learning, Computer Vision, Autonomous Navigation

EXPERIENCE

Teaching Assistant **June 2025 — September 2025**

University of Cape Town

- Act as the main point of contact between students and lecturers in the final year engineering professional communication course
- Support grading and lead tutorial sessions

Junior Computer Vision **Jan 2024 — Nov 2024**

Hawktivity

- Applied computer vision image processing (Gaussian filtering, Edge detection, Adaptive thresholding) and feature extraction (SIFT, SURF) methods with OpenCV in Python and scikit-image library to resolve challenges caused by non-static cameras with dirty lenses
- Optimized the YOLO model to detect specific objects in fast-food restaurants
- Conducted data annotation, accurately labeling objects to improve machine learning datasets specifically for fast-food restaurants
- Utilized AWS S3 for scalable data storage and retrieval of large datasets

Data Analyst Teaching Assistant **June 2023 — Dec 2023**

Edx/2u, remote

- Data Visualization: Tableau, PowerBi
- Database: SQL and API data modeling
- Teaching & Mentoring: Lesson planning, peer review, student support

Teaching Assistant, Part Time **June 2023 — Dec 2023**

University of Cape Town

- Guided engineering students through a comprehensive micromouse maze design project by providing hands-on technical support, troubleshooting design challenges in MATLAB and evaluating their project milestones
- Planned the 2023 UCT Open day event, showcasing research thesis projects within the UCT Electrical-Mechatronics Engineering Department

Internship **Jun 2022 — Jul 2022**

Alphawave Mobile Network Service

- CAD modeling for 3D designs

Internship

Nov 2021 — Dec 2021

Alphawave Mobile Network Service

- CAD modeling for 3D designs

PROJECTS

Game : Catch the Ball

- An interactive real-time animated UI game using Java AWT/Swing API.
- Concepts : OOP, file I/O, event-driven logic, and audio integration

E-Commerce Web Application

- An e-commerce web app featuring product listings, shopping cart, and checkout
- A user-friendly interface using HTML, CSS, and JavaScript

South Africa Weather Coordinated Web Page

- Develop an interactive web-based dashboard for real-time weather visualization and alerts across South African cities
- Integrate external weather APIs to display live weather data using geospatial visualizations
- Implement the frontend with Streamlit and PyDeck; design the backend using Flask (Python) and SQLite

Optimizing GPT-2 for South African Cuisine Chatbot

- Fine-tune GPT-2 on a domain-specific South Africa dataset
- Apply Reinforcement Learning from Human Feedback (RLHF) to align model responses with user intent
- Implement Retrieval-Augmented Generation (RAG) for improved contextual accuracy in dialogue generation